Nonlinear hydropower scheduling with coupled hydraulics and delayed river networks

Abstract

A hydropower schedule chooses when to run generating units and how much water to release. Each release changes both the available water and the head for subsequent generation; in a connected watercourse it also changes downstream supply after a travel delay. We formulate these decisions as a mixed-integer nonlinear optimization problem with reservoirs, shared tunnels, finite gate choices and river networks. Reservoir balances use midpoint storage to evaluate head. Pressure losses and efficiency-dependent power remain nonlinear. River transport accounts for both arrivals and water still in transit, with an explicit distinction between fixed travel times and distributed delays. Supplied curves are represented by their specified interpolants. Feasible schedules are found using discrete proposals and nonlinear dispatch; global search provides an upper bound on their possible improvement. Independent integration checks the returned controls on the optimization grid and a finer grid. The model and solution method are implemented in Julia with open-source solvers.

1 The scheduling problem

A reservoir's water can produce electricity now or be retained for later use. Releasing it changes the reservoir level, and therefore changes the energy available from the next release. In a cascade the same water may pass through several plants, with pressure losses along tunnels and travel delays along rivers. These effects connect decisions across the network and over time. Unit operation adds a discrete choice: an idle unit produces zero, whereas a running unit must remain within its operating range. Models combining these choices with head-dependent generation have a long history [1].

The given data are the network, initial volumes and operating histories, prices, inflows, equipment curves and operating limits. The decisions are unit states, unit discharges, outlet openings and tunnel positions chosen from finite sets. Reservoir volumes, tunnel flows, hydraulic heads, river arrivals and electrical output must then satisfy the equations below. The objective is revenue from generation, less operating penalties, plus the value assigned to water remaining at the end of the horizon.

We assume deterministic inputs and quasi-steady hydraulics: tunnel pressures and flows are in equilibrium within each scheduling interval, while reservoir storage changes over time. This excludes tunnel inertia and water hammer. River geometry enters through supplied discharge and travel-time response curves; the optimization does not solve river momentum equations. Pumping, reserve products and stochastic decisions are outside the present model.

The final formulation retains pressure losses, changing head and efficiency in its nonlinear equations. Linear approximations help propose initial discrete decisions. We first state the water balances, hydraulic laws and operating constraints, then explain how they are represented and solved. An exact representation of a supplied curve means the same interpolated function; the curve itself remains an approximation of the watercourse.

2 Water balance and hydraulic head

Let $0 = \tau_0 < \cdots < \tau_T$ be time edges in hours, $\Delta_t = \tau_{t+1} - \tau_t$ the interval duration, and $m_t = (\tau_t + \tau_{t+1})/2$ its midpoint, for $t = 0, ..., T-1$. Interval inputs are piecewise constant, with changes at these edges. Volumes are recorded at the edges; the hydraulic equations describe each interval using its midpoint reservoir volumes.

Hydraulic head is mechanical energy per unit weight, expressed in metres. At a reservoir it is the water-surface elevation. At a flooded junction it also includes pressure divided by ρg, where ρ is water density and g is gravitational acceleration. A head difference drives flow: some of that energy is dissipated by friction and some may be converted to electricity. Velocity corrections and local losses are included in the supplied equipment relationships rather than modeled as separate states.

We measure volume in Mm^3, here denoting one million cubic metres ($10^6\mathrm{m}^3$), discharge in $\frac{\mathrm{m}^3}{\mathrm{s}}$ and time in hours. The conversion factors are

$$\kappa = \frac{3600}{10^6} = 0.0036, \quad \gamma = \frac{\rho g}{10^6} = 0.00981 \\ \left(\rho = 1000\frac{\mathrm{kg}}{\mathrm{m}^3}, \quad g = 9.81\frac{\mathrm{m}}{\mathrm{s}^2}\right). \tag{1}$$

Thus $\kappa\Delta q$ converts a discharge over an interval into a volume, and $\gamma q h$ is hydraulic power at discharge q and net head h. For example, a release of $10\frac{\mathrm{m}^3}{\mathrm{s}}$ for one hour removes $0.036\mathrm{Mm}^3$ of water. A reservoir's volume change is this conversion applied to its net inflow.

To write the same accounting for a network, let $\mathcal{R}$, $\mathcal{J}$ and $\mathcal{B}$ index reservoirs, pressure junctions and prescribed-head boundaries. A node n is any of these three. Tunnels, instantaneous channels, generating units and delayed river reaches have index sets $\mathcal{E}$, $\mathcal{C}$, $\mathcal{G}$ and $\mathcal{D}$, respectively. The units at plant p form $\mathcal{G}_p$. Table 1 lists the variables. Underlines and overlines denote supplied lower and upper bounds.

Variable	Meaning	Unit
V_{it}, W_{rt}	Reservoir volume, river water in transit	Mm^3
H_{nt}, h_{pt}	Node head, turbine net head	m
Q_{et}, X_{ct}, q_{gt}	Tunnel, channel, unit discharge	$\frac{\mathrm{m}^3}{\mathrm{s}}$
R_{rt}, d_{rt}	Reach entry flow, reservoir withdrawal	$\frac{\mathrm{m}^3}{\mathrm{s}}$
A_{rt}	Volume arriving from a delayed reach	Mm^3
P_{gt}	Electrical power	MW
u_{gt}, U_{pt}	Unit and plant operating states	$\{0,1\}$
b_{et}, a_{rt}	Tunnel position, reservoir outlet opening	$[0,1]$

Table 1: Physical variables. Storage and river inventory use time vertices; other indexed quantities use control intervals.

Let F_{nt} be net flow into node n from the modeled connections, excluding natural reservoir inflow I_{it}. The matrices B^E, B^C and B^G record the tunnel, channel and unit connections. For a connection between nodes, its column is -1 at the source, $+1$ at the destination, and zero elsewhere. Negative discharge reverses these contributions. Arrival volumes from rivers and withdrawals into them give the remaining terms:

$$F_{nt} = \sum_e B^E_{ne} Q_{et} + \sum_c B^C_{nc} X_{ct} + \sum_g B^G_{ng} q_{gt} + \sum_r D_{nr}\frac{A_{rt}}{\kappa\Delta_t} - \sum_r S_{nr} d_{rt}. \tag{2}$$

$D_{nr} = 1$ when reach r delivers into node n, and $S_{nr} = 1$ when an outlet at n withdraws water into that reach; other entries are zero. Arrival volume A_{rt} is divided by $\kappa\Delta_t$ to give a flow rate. A tunnel or plant discharging into a river has its withdrawal in F, while its injection enters the river balance below. Its hydraulic outlet reference need not be its immediate water destination: a downstream reservoir can determine the plant's outlet head before the released water arrives there, as Figure 1 illustrates.

Water stored at the next time edge equals the present volume plus net inflow over the interval. A pressure junction stores no water, so its net inflow must be zero:

$$V_{i,t+1} - V_{it} = \kappa\Delta_t(I_{it} + F_{it}), \quad V_{i0} = V_i^{\mathrm{init}}. \tag{3}$$

$$F_{jt} = 0 \quad (j \in \mathcal{J}), \quad \underline{V}_{it} \le V_{it} \le \overline{V}_{it} \quad (t = 0, ..., T). \tag{4}$$

Initial volumes and volume limits are given data. Junction balances determine how flows meet and divide. Boundary nodes have prescribed heads and may exchange water with the exterior. Reservoir head depends on midpoint volume:

$$H_{it} = Z_i\left(\frac{V_{it} + V_{i,t+1}}{2}\right), \quad \underline{H}_j \le H_{jt} \le \overline{H}_j, \quad H_{bt} = H_b^{\mathrm{given}}. \tag{5}$$

Z_i converts volume to surface elevation. It is an increasing supplied table, or $Z_i(V) = z_i + s_i V + c_i V^2$ with given coefficients. Equations (3) and (5) form an implicit midpoint step: the ending volume helps determine the head that determines the interval flows. For nonlinear Z_i, applying the function to the average volume differs from averaging the two endpoint elevations.

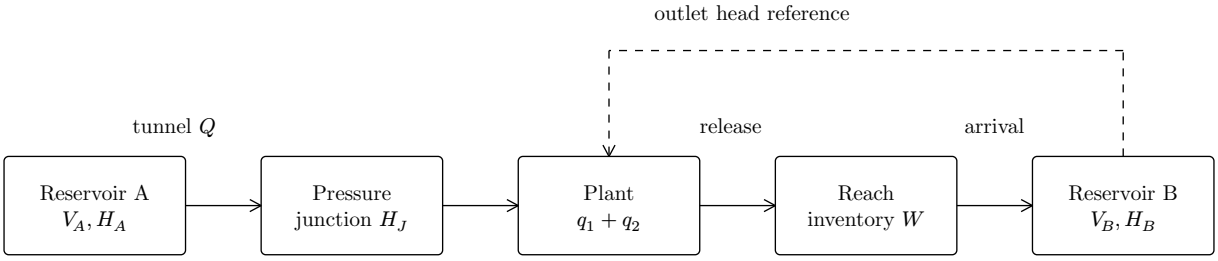


Figure 1: Water destination and hydraulic reference have different roles. Solid arrows move water; the dashed connection supplies the plant's outlet head. Water reaches B after transport, although its level already affects generation.

3 Tunnel hydraulics and generation

3.1 Tunnels and gates

For an oriented tunnel $e = (v, w)$, positive Q_e flows from v to w. With opening $b > 0$, the head difference equals friction loss:

$$H_{vt} - H_{wt} = K_e(b_{et})Q_{et}|Q_{et}|,$$
$$K_e(b) = \begin{cases} \frac{R_e}{b} & \text{without a gate-loss curve} \\ R_e + G_e(b) & \text{with a supplied gate-loss curve.} \end{cases} \tag{6}$$

$R_e > 0$ is tunnel resistance and $G_e \geq 0$ is any additional gate resistance; both have units $\frac{\mathrm{s}^2}{\mathrm{m}^5}$. For positive Q, upstream head exceeds downstream head; for negative Q, the difference reverses. The loss magnitude grows as Q^2, and energy is dissipated in the direction of flow. These head equations and the junction balances jointly determine flow sharing through branches and loops.

A closed gate permits a head difference while stopping flow:

$$b_{et} = 0 \Longrightarrow Q_{et} = 0, \quad |Q_{et}| \leq C_{et}. \tag{7}$$

Equation (6) therefore applies only to positive openings. Capacity C_{et} bounds flow in either direction; an outfall permitted to discharge only into a river also has $Q_{et} \geq 0$. Tunnel positions are prescribed or chosen from a finite supplied set $\mathcal{S}_e \subset [0, 1]$. A prescribed position may be fractional; arbitrary continuous tunnel positions are not optimization decisions in this model.

An instantaneous channel between reservoirs follows a supplied discharge law:

$$X_{ct} = \Phi_c\big(H_{vt} - H_{wt}, H_{\mathrm{ref},t}\big), \quad |X_{ct}| \leq C_{ct}. \tag{8}$$

Φ_c is a family of curves indexed by a specified upstream or downstream reservoir level. Each curve gives discharge as a function of level difference, is nondecreasing and passes through $(0, 0)$. Positive and negative responses need not be symmetric. We interpolate linearly along each curve and between reference levels. Outside the tabulated differences we extend the end segments; outside the reference levels we use the nearest curve. The channel transfers water in the same interval and has no separate volume in transit.

3.2 Energy conversion

Write $q_{pt} = \sum_{g \in \mathcal{G}_p} q_{gt}$ and $P_{pt} = \sum_{g \in \mathcal{G}_p} P_{gt}$ for plant totals. All units at a plant share the same net head:

$$h_{pt} = H_{\mathrm{source}(p),t} - \max\Big\{H_{\mathrm{target}(p),t}, h_p^{\mathrm{floor}}\Big\} - T_p\big(q_{pt}\big). \tag{9}$$

The target head is the outlet reference in Figure 1. A supplied floor prevents the effective outlet level from falling below h_p^{floor}; the maximum is omitted when no floor is given. $T_p\big(q_{pt}\big) \geq 0$ is the additional tailwater rise caused by total plant discharge. Shared penstocks are separate tunnels, so their friction depends on the combined flow of the units they supply.

The binary state u_{gt} selects an idle unit or its running range:

$$u_{gt} \in \{0, 1\}, \quad \underline{q}_{gt} u_{gt} \leq q_{gt} \leq \overline{q}_{gt} u_{gt}, \quad \underline{P}_{gt} u_{gt} \leq P_{gt} \leq \overline{P}_{gt} u_{gt}. \tag{10}$$

The electrical output is the available hydraulic power multiplied by turbine and electrical efficiencies:

$$P_{gt} = \gamma q_{gt} h_{pt} \eta_g\big(q_{gt}, h_{pt}\big) e_g\big(P_{gt}\big). \tag{11}$$

When $u_{gt} = 1$, the unit must also satisfy head and efficiency limits and, when supplied by a turbine table, head-dependent discharge limits:

$$\begin{gathered} \underline{h}_g \le h_{pt} \le \overline{h}_g, \quad \eta_g^{\min} \le \eta_g\big(q_{gt}, h_{pt}\big) \le 1, \\ q_g^{\mathrm{lo}}\big(h_{pt}\big) \le q_{gt} \le q_g^{\mathrm{hi}}\big(h_{pt}\big). \end{gathered} \tag{12}$$

When $u_{gt} = 0$, (10) gives $q = P = 0$ and the operating inequalities in (12) are inactive. The plant head is still determined by the hydraulic network, even when none of its units runs.

Turbine efficiency is a supplied table or the analytic relationship

$$\eta_g(q,h) = \eta_g^0 - \alpha_g \left(\frac{q - q_g^*}{q_g^*} \right)^2 - \beta_g \left(\frac{h - h_g^*}{h_g^*} \right)^2. \tag{13}$$

Here q_g^* and h_g^* are reference operating conditions, η_g^0 is their turbine efficiency, and nonnegative α_g, β_g describe its decrease away from them. Tables interpolate linearly in head and either linearly or with shape-preserving cubics in discharge.

Turbine efficiency converts hydraulic power to shaft power. Electrical efficiency e_g accounts for the remaining conversion to electrical output. It is a separate piecewise-linear function of *electrical output* P, so (11) must be solved for P even when q, h are known. Accepted curves make $P/e_g(P)$ strictly increasing, giving at most one admissible output for a given shaft power. With no electrical-efficiency curve, we use $e_g = 1$.

Plant state U_{pt} is one whenever at least one unit runs:

$$U_{pt} \ge u_{gt} \quad \big(g \in \mathcal{G}_p\big), \quad U_{pt} \le \sum_{g \in \mathcal{G}_p} u_{gt}, \quad U_{pt} \in \{0,1\}. \tag{14}$$

Plant limits apply to aggregate output and discharge: $P_{pt} \le \overline{P}_{pt}$ and $q_{pt} \le \overline{q}_{pt}$ when supplied, and $P_{pt} \ge \underline{P}_{pt} U_{pt}$, $q_{pt} \ge \underline{q}_{pt} U_{pt}$.

4 Outlets and river transport

4.1 Entry laws and confluences

A river *reach* is a link with a defined entry and downstream destination. For a reach fed by a reservoir outlet, total entry is $R_{rt} = n_{rt} + d_{rt}$: natural reach inflow n plus reservoir withdrawal d. With outlet opening a_{rt}, one chosen law determines withdrawal:

$$d_{rt} = \begin{cases} (C_{rt} - n_{rt}) a_{rt} & \text{controlled outlet} \\ c_r a_{rt} \sqrt{\ell_{rt}} & \text{orifice} \\ c_r \ell_{rt}^{\frac{3}{2}} & \text{uncontrolled weir} \\ a_{rt} D_r\big(H_{\mathrm{source}(r),t}\big) & \text{supplied discharge curve.} \end{cases} \tag{15}$$

ℓ is the head above the outlet crest, $H_{\mathrm{source}} - z_r^{\mathrm{crest}}$. It must be nonnegative unless dry operation is allowed, in which case we use $\big[H_{\mathrm{source}} - z_r^{\mathrm{crest}}\big]_+$, with $[x]_+ = \max(x, 0)$. The coefficient c_r or curve D_r is supplied. An uncontrolled weir has $a = 1$, also when its law is given as a table. Other openings satisfy their bounds in $[0, 1]$. Ordinary reaches obey $n_{rt} \le R_{rt} \le C_{rt}$. Only $d = R - n$ is subtracted from reservoir storage: the natural reach inflow enters the river directly and has never been in the reservoir.

At a confluence, water arriving from upstream reaches joins natural inflow and any plant or tunnel outfalls. Let L_{rt} collect these local injections:

$$L_{rt} = n_{rt} + \sum_{g \to r} q_{gt} + \sum_{e \to r} Q_{et}, \quad R_{rt} = L_{rt} + \sum_{s \to r} \frac{A_{st}}{\kappa \Delta_t}. \tag{16}$$

The notation $g \to r$, $e \to r$ and $s \to r$ identifies connections into reach r. Any number of upstream reaches may merge into one outgoing reach. No allocation decision is needed at such a junction. The delayed reach network is directed and acyclic, so upstream responses can be propagated downstream in order. The separate tunnel and instantaneous-channel network may contain loops.

4.2 Release cohorts and travel-time responses

A *cohort* is the water entering a reach during one release interval. With a fixed travel time, that release simply appears downstream later. With a distributed delay, different fractions of the cohort arrive at

different times. The distribution describes how water is spread in time; it does not introduce uncertainty into the scheduling problem. Supplied response functions take the place of a resolved river-flow model [2]. For reference entry flow ρ_{rj}, let fraction $w_{rj\ell}$ have a travel time uniformly distributed over $\left[\delta_{rj\ell}, \delta_{rj,\ell+1}\right]$. The delay edges are nonnegative and strictly increasing; $w_{rj\ell} \geq 0$ and $\sum_\ell w_{rj\ell} = 1$. The cumulative fraction that has arrived after elapsed time s is

$$F_{rj}(s) = \sum_\ell w_{rj\ell} \min\left\{1, \max\left\{0, \frac{s - \delta_{rj\ell}}{\delta_{rj,\ell+1} - \delta_{rj\ell}}\right\}\right\}. \tag{17}$$

Entry flow may change the travel response. For two or more nonnegative, strictly increasing reference flows covering the admissible entry range, interpolate between the neighboring distributions. In the following equations, SOS2 is a *special ordered set of type 2*: at most two weights may be nonzero, and their indices must be adjacent.

$$\theta_{rkj} \geq 0, \quad \sum_j \theta_{rkj} = 1, \quad R_{rk} = \sum_j \rho_{rj}\theta_{rkj}, \quad \theta_{rk} \in \mathrm{SOS2}\,. \tag{18}$$

The weights reproduce the cohort's total entry flow, including natural inflow and upstream arrivals. For example, halfway between two reference flows, the two distributions receive equal weights. With only one distribution, $\theta_{rk1} = 1$ and the flow-reproduction equation is omitted; the release can still vary. The selected mixture remains attached to the cohort after release. Changing an operating capacity does not change these reference flows.

Consider water entering at time s in interval k. The fraction of it arriving during interval t is the cumulative fraction at age $\tau_{t+1} - s$ minus that at age $\tau_t - s$. Averaging over the release interval gives

$$B_{rtkj} = \frac{1}{\Delta_k}\int_{\tau_k}^{\tau_{k+1}} \left[F_{rj}(\tau_{t+1} - s) - F_{rj}(\tau_t - s)\right] \mathrm{d}s. \tag{19}$$

Multiplying by released volume and summing over cohorts gives arrival volume A_{rt}:

$$A_{rt} = A_{rt}^{\mathrm{hist}} + \kappa \sum_k \Delta_k R_{rk} \sum_j \theta_{rkj} B_{rtkj}. \tag{20}$$

Water arriving at clock time s from cohort k must have a travel time between $s - \tau_{k+1}$ and $s - \tau_k$. Integrating the delay density over this range gives the difference of cumulative fractions below. Multiplying by the cohort's release rate gives its contribution to instantaneous arrival rate:

$$f_r(s) = f_r^{\mathrm{hist}}(s) + \sum_k R_{rk} \sum_j \theta_{rkj}\left[F_{rj}(s - \tau_k) - F_{rj}(s - \tau_{k+1})\right]. \tag{21}$$

The historical terms account for releases before τ_0. The integrals are exact for the supplied uniform bins. The interpolated quantity is the *fraction* that arrives. Multiplication by the flow-dependent released volume then makes the transfer piecewise quadratic in R. Interpolating released volumes directly would change this travel response.

When every reach has a fixed travel time $\delta_r \geq 0$, we retain the full piecewise-constant entry stream $\tilde{R}_r(s)$ instead of averaging it at each confluence. Shifting a stream by its delay and adding upstream arrivals gives

$$\begin{gathered} f_r(s) = \tilde{R}_r(s - \delta_r), \quad \tilde{R}_r(s) = L_r(s) + \sum_{s' \to r} f_{s'}(s), \\ A_{rt} = \kappa \int_{\tau_t}^{\tau_{t+1}} f_r(s)\,\mathrm{d}s. \end{gathered} \tag{22}$$

The confluence relation applies for $s \geq 0$. At a reservoir source, $\tilde{R}_r(s) = n_{rt} + d_{rt}$ on $[\tau_t, \tau_{t+1})$. For $s < 0$, it is the independently supplied entry history of that reach. Interval entry variables are the averages of these streams:

$$R_{rt} = \frac{1}{\Delta_t}\int_{\tau_t}^{\tau_{t+1}} \tilde{R}_r(s)\,\mathrm{d}s. \tag{23}$$

Keeping the stream preserves the time of every release change. For example, a one-hour release passing through two reaches with half-hour delays must arrive as a one-hour release shifted by one hour. On an hourly mixing grid, averaging after the first reach would spread some arrival before and after that final interval. Zero delays and delays shorter than a control interval allow newly released water to affect downstream flow in that same interval. Entry capacities are therefore also checked on the stream, including both sides of each jump.

If any reach has a distributed delay, (16) instead uses the mean upstream arrivals as the next reach's interval release. Equations (19) and (20) are exact for those rectangular cohorts, but this downstream averaging approximates the continuous timing. A finer transport grid tests its effect by comparing cumulative arrivals and water in transit. Conservation alone does not reveal a pulse that has arrived at the wrong time.

4.3 Water in transit, history and environmental flows

Water released but not yet delivered belongs to reach inventory W. Its change equals entry volume minus arrival volume:

$$W_{r,t+1} = W_{rt} + \kappa \Delta_t R_{rt} - A_{rt}, \quad W_{r0} = W_r^{\text{hist}}. \tag{24}$$

Pre-horizon entry records determine both initial inventory and future historical arrivals. For a past cohort k released on $[s_k, e_k)$, let F_k be its selected delay mixture. Water released at $s < 0$ has age $-s$ at the horizon start, so the fraction still in transit is $1 - F_k(-s)$:

$$W_r^{\text{hist}} = \kappa \sum_{k \in \mathcal{H}_r} R_{rk} \int_{s_k}^{e_k} [1 - F_k(-s)] \, \mathrm{d}s. \tag{25}$$

The set $\mathcal{H}_r$ contains the supplied historical cohorts of reach r; unspecified earlier entry is zero. Each reach's history already includes its past upstream contributions. Propagating those contributions again would count them twice. Restarting from an executed schedule therefore retains the original entry cohorts and their flows, along with reservoir states and operating histories. A past average arrival alone does not identify the release times needed to predict remaining arrivals.

Summing reservoir and reach balances cancels every internal transfer:

$$\begin{aligned} &\sum_i (V_{i,t+1} - V_{it}) + \sum_r (W_{r,t+1} - W_{rt}) \\ &\quad = \kappa \Delta_t \left(\sum_i I_{it} + \sum_r n_{rt} + E_t \right). \end{aligned} \tag{26}$$

Here E_t is net boundary exchange, positive into the modeled system. For a plant outfall, the source reservoir loses q, the reach receives q, and the receiving reservoir gains the later arrival A. These internal movements cancel; only natural inflows and external exchanges change total stored and in-transit water. The same identity includes reversed flow on instantaneous connections.

Environmental requirements can constrain reach entry $R_{rt} \geq R_{rt}^{\min}$, average arrival over a specified time window J, or arrival at every instant:

$$\int_J f_r(s) \, \mathrm{d}s \geq \int_J f_r^{\min}(s) \, \mathrm{d}s, \quad f_r(s) \geq f_r^{\min}(s). \tag{27}$$

The supplied policy chooses the window or pointwise arrival constraint. For rectangular cohorts, distributed arrival rates are piecewise linear; fixed-delay streams are piecewise constant. Between combined response and requirement breakpoints, the difference $f_r - f_r^{\min}$ is affine, so its minimum occurs at an endpoint. Endpoint checks thus enforce the pointwise constraint; jumps require both one-sided limits. At the horizon endpoints, use the limit from inside the horizon.

An aggregate flow requirement can also combine named unit discharges, river entries, channel flows and a given inflow contribution I_{ot}:

$$\sum_{g \in \mathcal{G}_o} q_{gt} + \sum_{r \in \mathcal{D}_o \cup \mathcal{C}_o} R_{rt} + I_{ot} \geq M_{ot}. \tag{28}$$

The sets $\mathcal{G}_o$, $\mathcal{D}_o$ and $\mathcal{C}_o$ select the contributions, and M_{ot} is the required minimum. For a channel, $R_{ct} = X_{ct}$ retains its sign. This constraint observes flows already accounted for in the network; it adds no water to any balance.

5 Operating constraints over time

A unit's initial operating state and its duration in that state are given. Binary starts s^+ and stops s^- record changes between consecutive intervals:

$$u_{gt} - u_{g,t-1} = s_{gt}^+ - s_{gt}^-, \quad s_{gt}^+ + s_{gt}^- \leq 1, \quad s_{gt}^{\pm} \in \{0, 1\}. \tag{29}$$

Minimum running time L_g^{up} means that a start at τ_t requires $u_{gk} \geq s_{gt}^{+}$ for each later interval with $0 \leq \tau_k - \tau_t < L_g^{\mathrm{up}}$. Minimum stopped time L_g^{down} similarly requires $u_{gk} \leq 1 - s_{gt}^{-}$ whenever $0 \leq \tau_k - \tau_t < L_g^{\mathrm{down}}$. These tests use elapsed hours, including on an irregular grid. Initial state age determines any remaining mandatory running or stopped time.

The same constraints apply to U when minimum plant operating times are specified. Replacing one running unit with another does not restart an already operating plant. An unfinished minimum duration at the horizon end is carried into a restart rather than imposed through an artificial terminal state.

Availability, forced states and prescribed output restrict the same decisions:

$$\underline{u}_{gt} \leq u_{gt} \leq \overline{u}_{gt}, \quad P_{gt} = P_{gt}^{\mathrm{given}}, \quad q_{gt} = q_{gt}^{\mathrm{given}}. \tag{30}$$

Only supplied schedules are imposed. Plant state also satisfies $\underline{U}_{pt} \leq U_{pt} \leq \overline{U}_{pt}$, and plant schedules constrain the corresponding sums. Maintenance forces units off; zero and positive output schedules require the appropriate off and on states. All restrictions, including initial minimum durations, apply together. Inconsistent restrictions make the model infeasible.

For interval-average discharge or release x_t, supplied upward and downward rates r_x^{+}, r_x^{-} limit changes between interval midpoints:

$$-\int_{m_{t-1}}^{m_t} r_x^{-}(s)\,\mathrm{d}s \leq x_t - x_{t-1} \leq \int_{m_{t-1}}^{m_t} r_x^{+}(s)\,\mathrm{d}s. \tag{31}$$

A running unit cannot enter its positive operating range from zero when the allowed inter-interval increase is smaller than its minimum output. Power ramps therefore allow the minimum-output jump at startup and shutdown: add $\sum_{g \in \mathcal{G}_x} \underline{P}_{gt} s_{gt}^{+}$ to the upward allowance and $\sum_{g \in \mathcal{G}_x} \underline{P}_{g,t-1} s_{gt}^{-}$ to the downward allowance. Here $\mathcal{G}_x$ contains that unit or all units at the constrained plant. Discharge and river-release ramps have no jump allowance. The first interval uses a supplied previous window's average and midpoint; if absent, only that comparison is omitted.

Reservoir volume and level ramps instead constrain vertex changes $V_{i,t+1} - V_{it}$ and $Z_i\big(V_{i,t+1}\big) - Z_i(V_{it})$, with rate integrals over $[\tau_t, \tau_{t+1}]$. These bounds limit changes between recorded states, not derivatives at every instant.

A tunnel interlock limits how many gates in a group $\mathcal{E}_\ell$ may be open at once. With $o_{et} = \mathbf{1}[b_{et} > 0]$,

$$\underline{N}_{\ell t} \leq \sum_{e \in \mathcal{E}_\ell} o_{et} \leq \overline{N}_{\ell t}. \tag{32}$$

A positive position counts as open even when its flow is zero.

6 Economic objective

Generation at price π_t per MWh earns $\pi_t \Delta_t P_{gt}$. Let c^{+}, c^{-} be unit transition costs, c^{b} nonnegative gate-movement costs, c^{r} nonnegative release-shortfall costs, and v_i, v_r the values assigned to water in reservoirs and reaches, per Mm^3. We maximize

$$\begin{aligned} \mathcal{V} = &\sum_t \pi_t \Delta_t \sum_g P_{gt} - \sum_{g,t} \big(c_{gt}^{+} s_{gt}^{+} + c_{gt}^{-} s_{gt}^{-}\big) \\ &- \sum_{e,t} c_{et}^{b} \big| b_{et} - b_{e,t-1} \big| - \sum_{r,t} c_{rt}^{r} \xi_{rt} \\ &+ \sum_i v_i (V_{iT} - V_{i0}) + \sum_r v_r (W_{rT} - W_{r0}). \end{aligned} \tag{33}$$

Starts, stops and gate movements are charged once per event, without a duration multiplier. The first gate comparison uses its supplied initial position. The last two terms value changes in reservoir storage and reach inventory. Valuing water in transit allows a late release to retain downstream economic value even when it has not yet arrived. These terminal values are input assumptions about use beyond the horizon.

A positive release penalty permits a deficit below the entry minimum. The deficit variable is measured as a volume:

$$\xi_{rt} \geq 0, \quad \xi_{rt} \geq \kappa \Delta_t (R_{rt}^{\min} - R_{rt}). \tag{34}$$

A positive cost makes ξ equal the actual shortfall at optimum. Without that penalty, the entry minimum is hard and $\xi = 0$. Arrival minima and aggregate flow requirements remain hard. Subtracting initial water inventories in (33) removes a constant endowment from the reported value while leaving the maximizing decisions unchanged.

7 Representing curves and strengthening bounds

Together, the preceding equations and inequalities define a mixed-integer nonlinear program (MINLP). The head, efficiency and travel-response couplings are nonconvex even when unit states and gate choices are fixed. We distinguish two tasks: representing the supplied functions without changing them, and constructing relaxations that bound the objective during global search.

7.1 Representing supplied curves exactly

Direct evaluation of a curve is convenient for nonlinear dispatch. Global search also needs its inputs and outputs exposed as algebraic variables. The *graph* of a function is simply the set of input-output pairs that obey it. For a piecewise-linear curve with knots (x_i, y_i), its graph is

$$\lambda_i \ge 0, \quad \sum_i \lambda_i = 1, \quad x = \sum_i x_i \lambda_i, \quad y = \sum_i y_i \lambda_i, \quad \lambda \in \mathrm{SOS2}\,. \tag{35}$$

The SOS2 condition selects one segment [3]. Its two neighboring weights sum to one, and their weighted input coordinate fixes the interpolation fraction. The weighted output therefore gives exactly the supplied linear interpolant. Allowing nonadjacent weights would also allow points between distant knots and would change the curve.

For a turbine table, discharge weights λ_i and head weights μ_j each satisfy the corresponding SOS2 and coordinate constraints. Let E_{ij} be efficiencies at the knots. Linear interpolation in head combines the values η_j on the neighboring head curves. A shape-preserving cubic in discharge [4] is represented by adding corrections to the linear curve:

$$\begin{gathered} w_i = \lambda_i \lambda_{i+1}, \quad r_i = \lambda_i w_i, \\ \eta_j = \sum_i E_{ij}\lambda_i + \sum_i \big[A_{ij} r_i + B_{ij}(w_i - r_i)\big], \quad \eta = \sum_j \mu_j \eta_j. \end{gathered} \tag{36}$$

For segment width $\delta q_i = q_{i+1} - q_i$, secant slope $d_{ij} = \big(E_{i+1,j} - E_{ij}\big)/\delta q_i$ and original endpoint derivatives m_{ij}, use

$$A_{ij} = \delta q_i \big(m_{ij} - d_{ij}\big), \quad B_{ij} = \delta q_i \big(d_{ij} - m_{i+1,j}\big). \tag{37}$$

On the active segment, the two correction terms are multiples of $\lambda_i^2 \lambda_{i+1}$ and $\lambda_i \lambda_{i+1}^2$. Both vanish at the endpoints; their coefficients make the endpoint derivatives equal the supplied ones. The auxiliary products w_i, r_i implement these terms using quadratic equalities. Setting $A = B = 0$ gives bilinear interpolation. Restricting a domain does not refit the original cubic slopes. Outside the discharge table we use the declared end secants; head extrapolation, when allowed, uses its outer secant. An idle unit's network head may lie outside the head range of its running table. A finite extension keeps the algebraic relation defined there; $q = P = 0$, and the running head and efficiency limits remain inactive.

Units sharing the same net-head variable and compatible table grids can reuse head weights. A release cohort likewise uses one set of flow weights for its arrivals, environmental checks and terminal inventory, ensuring one travel response throughout. Channel responses use the analogous piecewise-bilinear graph. The product equalities in (36), (11) and (20) remain nonlinear: these representations preserve the curves rather than turning the model into a MILP.

7.2 Quadratic equations for tunnels and gates

A binary direction variable separates positive and negative tunnel flow:

$$\begin{gathered} Q = Q^+ - Q^-, \quad 0 \le Q^+ \le Cz, \quad 0 \le Q^- \le C(1-z), \\ S = (Q^+)^2 - (Q^-)^2, \quad z \in \{0,1\}. \end{gathered} \tag{38}$$

If $z = 1$, only Q^+ can be positive; if $z = 0$, only Q^- can be positive. Thus $S = Q|Q|$ in both cases. A directed outfall needs only $S = Q^2$; known head ordering can also fix a direction.

For each finite gate position, a binary selector identifies whether that position is chosen. Exactly one selector is active, so b, the open state o and the selected resistance $K(b)$ are weighted sums of known values. To impose the loss equation only when the gate is open, define $d_o = o(H_v - H_w)$. If $L \le H_v - H_w \le U$, this binary product is represented by

$$Lo \le d_o \le Uo, \quad H_v - H_w - U(1-o) \le d_o \le H_v - H_w - L(1-o). \tag{39}$$

At $o = 0$ these inequalities force $d_o = 0$; at $o = 1$ they force $d_o = H_v - H_w$. Together with $|Q| \le Co$ and $d_o = K(b)S$, they recover (6) and (7). Any finite resistance may be assigned to the closed selector because $S = 0$ there. This construction needs only quadratic product equalities. Similarly, a wet/dry

binary branch represents $\ell = \max(H - z^{\text{crest}}, 0)$, and a nonnegative auxiliary root r with $r^2 = \ell$ represents the square root in the outlet law (15).

7.3 Upper bounds for global optimization

A *relaxation* allows additional combinations of variables while retaining every feasible schedule. Its maximum objective is therefore an upper bound on what the physical model can earn. Relaxed flow, head and efficiency may combine to predict physically impossible power. A *valid inequality* excludes some of these combinations without excluding a feasible schedule; it tightens the bound while leaving the physical equations unchanged.

Finite storage and equipment limits provide variable domains. Conservation and inventory propagation can narrow them using the initial and terminal restrictions. These bounds cover all admissible schedules, rather than a neighborhood of one previously found solution.

For clarity, omit unit and time indices in the following bounds. For a running unit with nonnegative q, h, η, let $\overline{e}$ bound its electrical efficiency. Then $p^{\text{ub}}(q,h) = \gamma \overline{e} q h \eta(q,h)$ is an upper bound on physical electrical power. It is polynomial within each interpolation cell even when (11) is implicit in P. Suppose a plane $aq + bh + c$ bounds this polynomial throughout the running domain, with $b \geq 0$. For a head lower bound h_* valid in both operating states, the following inequality includes the idle state:

$$P \leq aq + b(h - h_*) + (c + bh_*)u. \tag{40}$$

When the unit runs, it reduces to $P \leq aq + bh + c$. When it is idle, $P = q = 0$ and the right side is nonnegative. To establish c, we bound the residual $p^{\text{ub}} - aq - bh$ over every original interpolation cell. In a Bernstein representation of a polynomial on a box, the basis weights are nonnegative and sum to one, so its value cannot exceed its largest coefficient. This supplies a bound over the entire cell, including interior extrema that a plane fitted at a few samples might miss.

Another relaxation retains joint information about flow, head and efficiency. Consider a box containing their admissible running values, with up to eight corners (q_c, h_c, η_c). Exact binary-product inequalities, as in (39), define $h_{\text{on}} = uh$ and $\eta_{\text{on}} = u\eta$. Nonnegative weights z_c on the corners satisfy

$$\begin{gathered} \sum_c z_c = u, \quad \sum_c q_c z_c = q, \quad \sum_c h_c z_c = h_{\text{on}}, \quad \sum_c \eta_c z_c = \eta_{\text{on}}, \\ \gamma e_{\min} \sum_c q_c h_c \eta_c z_c \leq P \leq \gamma e_{\max} \sum_c q_c h_c \eta_c z_c. \end{gathered} \tag{41}$$

Here $e_{\min}, e_{\max}$ bound electrical efficiency over the power range. For any point in the box, multilinear interpolation weights reproduce its coordinates and the product $qh\eta$. Hence the inequalities include every physical running point. When $u = 0$, all corner weights vanish. The original efficiency and power equalities still determine the physical output.

For a tabulated turbine with head knots h_j, head weights μ_j already available in its graph can also strengthen power bounds. We introduce ζ_j to equal the head weights while running and vanish while idle:

$$\begin{gathered} 0 \leq \zeta_j \leq \mu_j, \quad \sum_j \zeta_j = u, \quad \sum_j (h_j - h_1)\zeta_j = h_{\text{on}} - h_1 u, \\ P \leq aq + \sum_j c_j \zeta_j. \end{gathered} \tag{42}$$

The coefficients c_j are chosen so that the weighted expression bounds the polynomial upper bound p^{ub}, including its variation between head knots. The constraints give $\zeta = \mu$ when $u = 1$, and $\zeta = 0$ when $u = 0$ without fixing the idle hydraulic head. Discharge weights support analogous bounds along the other table axis. Additional planes may be generated during the search, provided each is checked over its complete domain. All these inequalities strengthen the relaxation while retaining the original equations.

8 Solving the model and checking a schedule

8.1 Finding feasible controls

JuMP provides the algebraic interface in Julia [5]. SCIP is used for both linear mixed-integer proposals and global nonlinear search [6]; Ipopt solves continuous nonlinear dispatch with discrete choices fixed [7]. The preparatory step uses a MILP approximation to propose unit states and finite gate positions. Nonlinear dispatch then adjusts continuous decisions under the original physical laws for those choices. Several proposals can be tried within a shared time allowance. Candidates passing the independent checks below are submitted as starts for the full SCIP model, where the discrete decisions are free again.

This preparation seeks feasible schedules, not bounds on the best possible objective. Its linear objective is a surrogate for (33). Dispatch with these choices fixed remains a piecewise-smooth, nonconvex nonlinear program, so convergence to a stationary point does not establish global optimality.

8.2 Reconstructing the water trajectory

An independent integration holds the selected unit discharges and openings fixed and recovers the resulting water trajectory. At each interval, Newton's method solves jointly for ending storage, junction heads, tunnel flows and active outlet releases. With a delay shorter than the interval, some current release arrives before the interval ends; that arrival must participate in the same solve rather than being computed only from past releases. Electrical output is then recovered from (11).

Some isolated pressure components determine only head differences. Adding the same constant to all their heads leaves the flow equations unchanged. One numerical reference removes this indeterminacy during solution, while every water balance is retained. The recovered heads can be shifted together to respect the junction bounds when such a shift is admissible.

Acceptance first requires the reconstructed schedule to satisfy the original interval equations and operating restrictions. Integration is then repeated on a finer grid with the same controls. Schedules and ramps are still checked over their original operating windows; storage at the original vertices, total water, cumulative arrivals and terminal inventory are compared separately. For supplied volume and relative objective tolerances $\varepsilon_V, \varepsilon_{\mathcal{V}}$, require also

$$\begin{gathered}\max_{i,t}\left|V_i^{\mathrm{fine}}(\tau_t) - V_{it}\right| \le \varepsilon_V, \quad \max_{r}\left|W_r^{\mathrm{fine}}(\tau_T) - W_{rT}\right| \le \varepsilon_V, \\ \left|\mathcal{V}^{\mathrm{fine}} - \mathcal{V}\right| \le \varepsilon_{\mathcal{V}} \max(1, |\mathcal{V}|).\end{gathered} \tag{43}$$

This refinement tests numerical consistency with the original controls. It estimates the effect of the time grid rather than rigorously bounding continuous-time hydraulic error.

8.3 Bounding the remaining objective improvement

A feasible schedule gives an achievable objective value L. Global search provides an upper bound U that no feasible schedule in the declared model can exceed. SCIP obtains such bounds from relaxations, tightens them and branches on discrete choices or continuous variable ranges. A region can be discarded when its bound cannot improve the current feasible value. A good schedule therefore helps the search even before it is proved optimal.

Here L is the independently reconstructed value of a schedule feasible for the interval model, and U must be a finite enclosing bound. The absolute and relative limits on possible improvement are

$$g_{\mathrm{abs}} = \max(0, U - L), \quad g_{\mathrm{rel}} = \frac{\max(0, U - L)}{\max(1, |L|)}. \tag{44}$$

For example, $L = 100$ and $U = 103$ in the same monetary units limit improvement to at most 3, or 3% under this relative-gap definition. They do not imply that a better schedule exists. Subtracting the initial water endowment in (33) leaves maximizing decisions unchanged but affects a relative gap's denominator; both L and U use that same objective convention.

A certificate records that the requested absolute or relative gap has been met for the chosen interval model, interpolation rules and finite gate sets. If discrete choices are fixed, it is conditional on those choices. The finer-grid check answers a separate question about the physical trajectory: a schedule can have a certified interval-model gap and still fail finer integration. Neither result certifies a continuous-time optimum. These are floating-point numerical certificates, interpreted at solver tolerances. They do not include a rigorous enclosure of every rounding error in the solver and polynomial bound calculations.

9 Discussion

The formulation connects storage, pressure loss, generation and travel time within one scheduling problem. Head and efficiency change as decisions change, rather than being fixed beforehand in a production approximation. River inventory gives released water an explicit place in the balance until it arrives. Supplied curves remain the functions chosen by the modeler, including their stated interpolation and extension rules.

The solution method uses this same physical model in distinct ways. Nonlinear dispatch finds feasible controls; independent integration checks what those controls do; global bounds limit the economic improvement left unresolved. Stronger relaxations help the global search without replacing hydraulic laws. These choices preserve the model's couplings, but they also preserve its nonconvexity. A large network may still require substantial time to close the optimality gap.

The numerical model has a stated time resolution, river response approximation and finite set of tunnel positions. Calibration, transient hydraulic regimes and uncertain future inputs remain separate modeling questions. Extensions can add explicit physical laws and operating constraints while using the same water accounting and distinguishing feasible operation from objective bounds and time-grid consistency.

References

[1] J. P. S. Catalão, H. M. I. Pousinho, and V. M. F. Mendes, "Mixed-integer nonlinear approach for the optimal scheduling of a head-dependent hydro chain," *Electric Power Systems Research*, vol. 80, no. 8, pp. 935–942, 2010, doi: 10.1016/j.epsr.2009.12.015.

[2] J. C. I. Dooge, "A general theory of the unit hydrograph," *Journal of Geophysical Research*, vol. 64, no. 2, pp. 241–256, 1959, doi: 10.1029/JZ064i002p00241.

[3] J. Huchette and J. P. Vielma, "A combinatorial approach for small and strong formulations of disjunctive constraints," *Mathematics of Operations Research*, vol. 44, no. 3, pp. 793–820, 2019, doi: 10.1287/moor.2018.0946.

[4] F. N. Fritsch and J. Butland, "A method for constructing local monotone piecewise cubic interpolants," *SIAM Journal on Scientific and Statistical Computing*, vol. 5, no. 2, pp. 300–304, 1984, doi: 10.1137/0905021.

[5] I. Dunning, J. Huchette, and M. Lubin, "JuMP: A modeling language for mathematical optimization," *SIAM Review*, vol. 59, no. 2, pp. 295–320, 2017, doi: 10.1137/15M1020575.

[6] S. Vigerske and A. Gleixner, "SCIP: global optimization of mixed-integer nonlinear programs in a branch-and-cut framework," *Optimization Methods and Software*, vol. 33, no. 3, pp. 563–593, 2018, doi: 10.1080/10556788.2017.1335312.

[7] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical Programming*, vol. 106, pp. 25–57, 2006, doi: 10.1007/s10107-004-0559-y.